

XIAOKUN (KEN) ZHONG

✉ xzhongal@berkeley.edu  github.com/Kenxzk  linkedin.com/in/xiaokun-ken-zhong

Work Authorization: U.S. Permanent Resident (I-485 filed, EB-5). No visa sponsorship required.

EDUCATION

University of California, Berkeley

Dec 2025 – May 2026

Visiting Student, Computer Science & Mathematics

Berkeley, CA

Hong Kong University of Science & Technology (HKUST)

Sep 2022 – Jun 2026

BSc in Mathematics (Computer Science Track)

Hong Kong

- Coursework: Differential Equations, Artificial Intelligence, Probability Theory.

RESEARCH & ENGINEERING EXPERIENCE

Dartmouth College & UC Berkeley

Feb 2025 – Present

Research Engineer, AI Systems & Scientific ML

Remote / Berkeley, CA

- Investigated failure modes and training instability in scientific AI systems under challenging regimes (e.g., stiff PDE settings), identifying factors affecting convergence and generalization.
- Designed and implemented second-order optimization pipelines (Newton-CG) in PyTorch, improving solution accuracy by up to 50% and enabling stable training across benchmark tasks.
- Built scalable ML experimentation infrastructure on Slurm-based HPC clusters, enabling large-scale ablation studies and reproducible evaluation of optimization strategies.
- Led research on regime-specific optimization for scientific ML, resulting in a co-first author submission to ICML 2026.

HKUST Undergraduate Research Program

Feb 2023 – May 2025

Research Engineer, Medical Imaging AI

Hong Kong

- Developed deep learning models for MRI reconstruction and 3D volumetric modeling from sparse medical data.
- Improved data processing and training pipelines in Python and C++, enabling more efficient experimentation on large-scale imaging datasets.

PUBLICATIONS

Y. Wang*, Y. Hu*, **X. Zhong*** et al. “Unveiling Multi-regime Patterns in SciML: Distinct Failure Modes and Regime-specific Optimization.” **ICML 2026**. arXiv:2605.29153.

TECHNICAL SKILLS

Machine Learning: PyTorch, scientific machine learning (PINNs, Neural ODEs), optimization, training dynamics

Systems & Infrastructure: Linux, Slurm, Docker, HPC workflows, distributed experiments

Languages: Python, C++, SQL, MATLAB

SELECTED PROJECTS

Secure Network Tunneling & Transport Optimization

Nov 2025

Systems Project

Remote

- Designed and deployed a secure cross-region networking system on VPS infrastructure to improve reliability and latency under constrained conditions.
- Implemented TLS-based transport and system-level optimizations, analyzing trade-offs between throughput, latency, and network stability.

Autonomous Robotics Vision System (RoboMaster)

Oct 2022 – Jun 2023

Computer Vision Engineer

Hong Kong

- Built real-time computer vision pipelines for autonomous target acquisition using OpenCV and Python.
- Implemented C++ drivers for sensor fusion and motor control on embedded STM32 platforms.

INDUSTRY EXPERIENCE

Agricultural Bank of China

Jan 2024 – Feb 2024

IT Intern (e-CNY Digital Currency)

Shenzhen, China

- Tested and improved security and functionality of an internal e-CNY wallet system, contributing to a large-scale digital currency pilot.